Noisy Data

- **Noise:** random error or variance in a measured variable
- **Incorrect attribute values** may be due to
  - Faulty data collection instruments
  - Data entry problems
  - Data transmission problems
  - Technology limitation
  - Inconsistency in naming convention
- **Other data problems**
  - Duplicate records
  - Incomplete data
  - Inconsistent data

How to Handle Noisy Data?

- **Binning**
  - First sort data and partition into (equal-frequency) bins
  - Then one can **smooth by bin means, smooth by bin median, smooth by bin boundaries**, etc.
- **Regression**
  - Smooth by fitting the data into regression functions
- **Clustering**
  - Detect and remove outliers
- **Semi-supervised: Combined computer and human inspection**
  - Detect suspicious values and check by human (e.g., deal with possible outliers)